ig.degree.betweenness: Detecting Communities When Order and Direction Matter in Social Network Analysis

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Background: There are limited algorithms for understanding communities in directed graphs. A novel algorithm combining both degree and betweenness centrality (ig.degree.betweenness) was developed to highlight gradients in connectivity. It identifies communities within complex directed graphs. These are networks that are comprised of actors who have ties that originate and end at themselves (self-loops).

Objectives: Hierarchies in social partitioning for degree centrality (in-ties, out-ties) is evident among actors when applying the ig.degree.betweenness algorithm to social networks. This characteristic has useful application in epidemiological research.

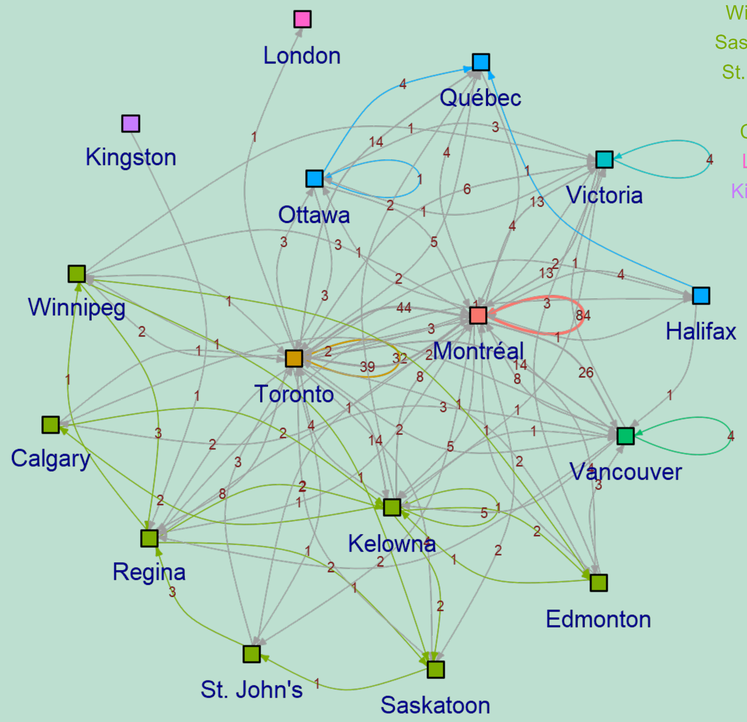
Methods: A simulated dataset of an infectious disease outbreak with its conjectured epicentre in Montréal is studied. Both the egocentric and sociocentric perspectives are investigated in a contact tracing approach among Canadian localities.

Results: In the egocentric perspective, 362 individuals with a suspect infection resided at least one night in a facility tied to outbreak during an event week. Of these individuals, 188 relocated to 14 other Canadian localities, and 174 relocated within Montréal. In the sociocentric perspective that encompasses Canada, there were 389 suspect cases with 470 relocations among 16 localities. Application of the ig.degree.betweenness algorithm yielded eight contact tracing communities among these Canadian localities. A gradient was evident in connectivity for degree centrality among the communities.

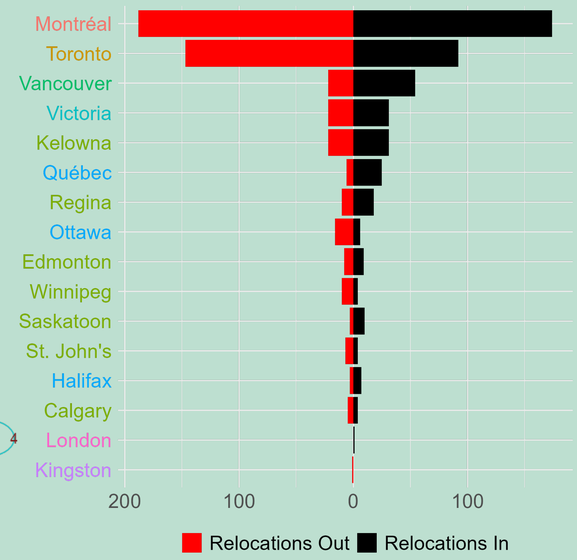
Conclusions: Community detection algorithms have been underutilized in epidemiological research. They provide insight on latent structures present in social networks. Our new algorithm reflects gradients in connectivity for complex directed graphs.

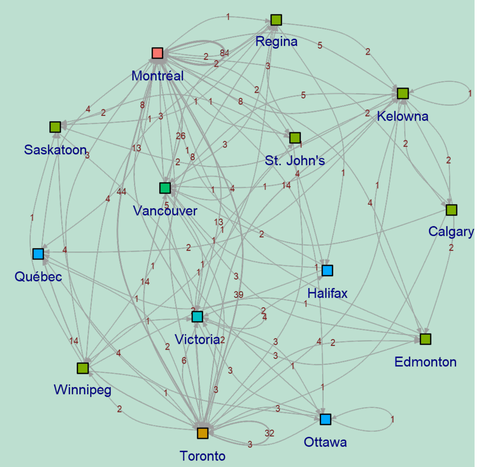
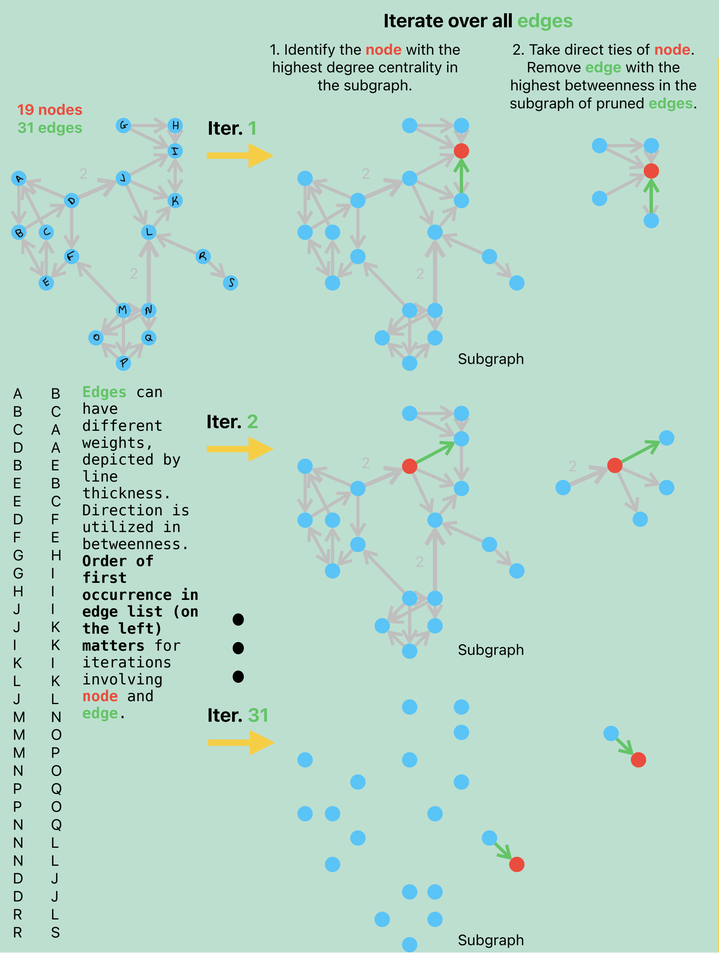
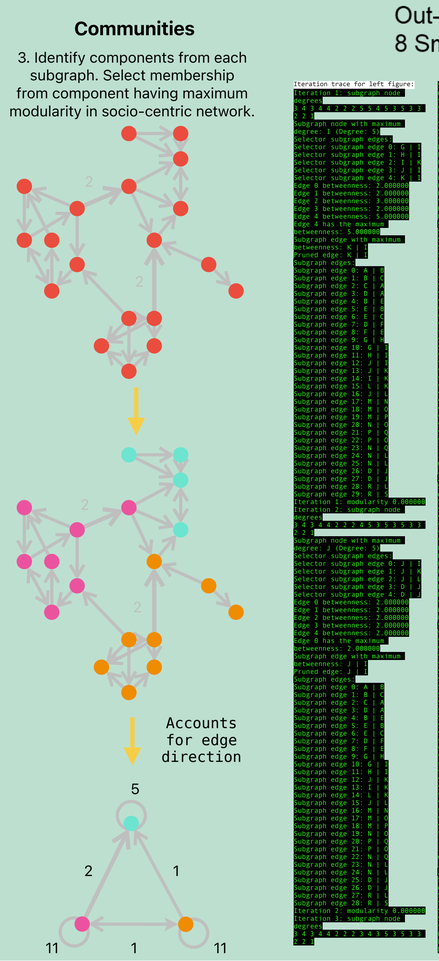
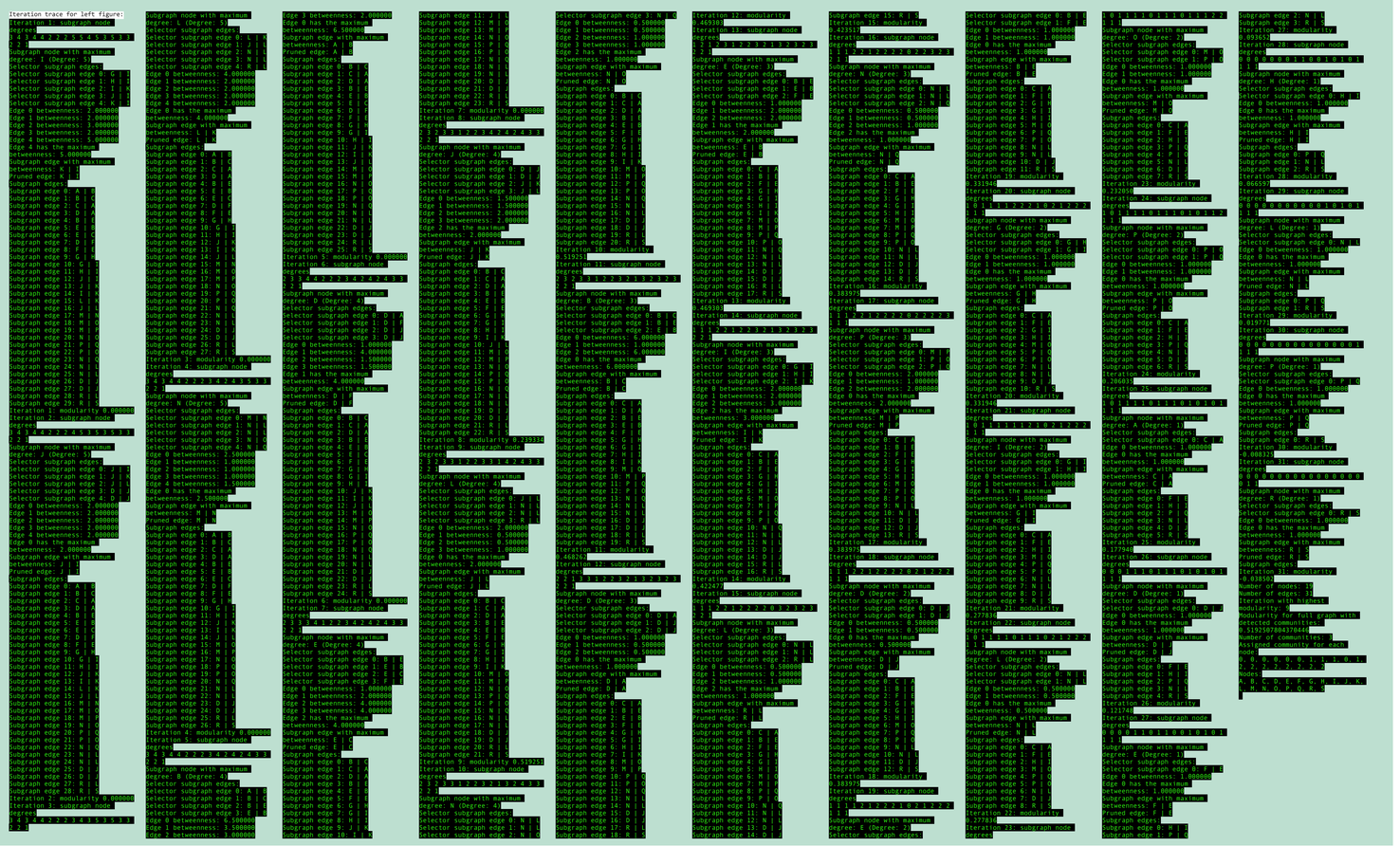
Visit our Github page at:
https://github.com/benyamindsmith/ig.degree.betweenness

Download latest R package from CRAN at:
https://cran.r-project.org/package=ig.degree.betweenness



Nodes: 16 localities, colour denotes community detection
Edges: 470 residence relocations, label denotes individual movement
Suspect cases by locality at one point over the week: 19.5 (1, 362)
In-degree centrality of suspect cases by locality: 9.5 (0, 174)
Out-degree centrality of suspect cases by locality: 9 (0, 188)
8 Smith-Pittman cluster communities among 16 localities with modularity of 0.08





Nodes: 14 localities in ego network for Montréal
Edges: 468 residence relocations, label denotes individual movement
Suspect cases by locality at one point over the week: 25 (9, 362)
174 relocations in, and 188 relocations out for Montréal